(一) 拉格朗日方程的推导：

假设：质点系包含 n 个质点，由于受到 r 个约束，自由度 $s = 3n-r$，选择 s 个广义坐标：

① 主动力：$F = (F_1, F_2, \cdots, F_n)$

② 虚位移：$\delta r = (\delta r_1, \delta r_2, \cdots, \delta r_n)$

③ 广义坐标：$q = (q_1, q_2, \cdots, q_n)$

④ 坐标变换方程：$r_i = r_i(q_1, q_2, \cdots, q_s, t)$

1. 首先证明几个有用的关系式：

$$\dot{\vec{r}}_i = \frac{d}{dt}\vec{r}_i(q_1, q_2, \cdots, q_s, t) = \sum_{k=1}^{s}\frac{\partial \vec{r}_i}{\partial q_k}\dot{q}_k + \frac{\partial \vec{r}_i}{\partial t} = \vec{r}_i(q_1, q_2, \cdots, q_s, \dot{q}_1, \dot{q}_2, \cdots, \dot{q}_s, t)$$

(1)
$$\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_k} = \frac{\partial}{\partial \dot{q}_k}\left(\sum_m \frac{\partial \vec{r}_i}{\partial q_m}\dot{q}_m + \frac{\partial \vec{r}_i}{\partial t}\right) = \frac{\partial \vec{r}_i}{\partial q_k}$$

$$\Rightarrow \boxed{\frac{\partial \dot{\vec{r}}}{\partial \dot{q}_k} = \frac{\partial \vec{r}_i}{\partial q_k}} \quad \text{关系1}$$

说明：r_i，$\frac{\partial r_i}{\partial t}$，$\frac{\partial r_i}{\partial q_k}$ 都只是 (t, q) 的函数，与广义速度 $\dot{q}$ 无关，在方程没有求解出来前 q，$\dot{q}$ 都是独立的.

(2) $$\frac{\partial \dot{\vec{r}}_i}{\partial q_k} = \frac{\partial}{\partial q_k}\left(\frac{d\vec{r}_i}{dt}\right) = \frac{\partial}{\partial q_k}\left(\sum_m \frac{\partial \vec{r}_i}{\partial q_m}\dot{q}_m + \frac{\partial \vec{r}_i}{\partial t}\right) = \sum_m \frac{\partial^2 \vec{r}_i}{\partial q_k \partial q_m}\dot{q}_m + \frac{\partial^2 \vec{r}_i}{\partial q_k \partial t}$$

$$= \sum_m \dot{q}_m \frac{\partial^2 \vec{r}_i}{\partial q_m \partial q_k} + \frac{\partial^2 \vec{r}_i}{\partial t \partial q_k} = \left[\sum_m \dot{q}_m \frac{\partial}{\partial q_m} + \frac{\partial}{\partial t}\right]\left(\frac{\partial \vec{r}_i}{\partial q_k}\right) = \frac{d}{dt}\left(\frac{\partial \vec{r}_i}{\partial q_k}\right)$$

$$\Rightarrow \quad \boxed{\frac{\partial \dot{\vec{r}}_i}{\partial q_k} = \frac{d}{dt}\left(\frac{\partial \vec{r}_i}{\partial q_k}\right)} \quad \text{关系二}$$

上式表明：时间全微分算符 $\frac{d}{dt}$ 与对广义坐标的偏导数算符 $\frac{\partial}{\partial q_k}$ 对易，即：

$$\frac{\partial}{\partial q_k}\frac{d}{dt} = \frac{d}{dt}\frac{\partial}{\partial q_k}$$

(3) 应用关于体系动能的概念：

$$T = \sum_{i=1}^{n} \frac{1}{2} m_i \dot{\vec{r}}_i^{\,2} = T(q_1, q_2, \cdots, q_s, \dot{q}_1, \dot{q}_2, \cdots, \dot{q}_s, t)$$

$$\Rightarrow \quad \boxed{\begin{cases} \dfrac{\partial T}{\partial q_k} = \displaystyle\sum_{i=1}^{n} \frac{\partial T}{\partial \dot{\vec{r}}_i} \cdot \frac{\partial \dot{\vec{r}}_i}{\partial q_k} = \sum_{i=1}^{n} m_i \dot{\vec{r}}_i \cdot \frac{\partial \dot{\vec{r}}_i}{\partial q_k} \\ \dfrac{\partial T}{\partial \dot{q}_k} = \displaystyle\sum_{i=1}^{n} \frac{\partial T}{\partial \dot{\vec{r}}_i} \cdot \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_k} = \sum_{i=1}^{n} m_i \dot{\vec{r}}_i \cdot \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_k} \end{cases}} \quad \text{关系三.}$$

2. 下面由达朗贝尔-拉格朗日方程(动力学普遍方程)推导拉格朗日方程：

$$\sum_i (\vec{F}_i - m_i\ddot{\vec{r}}_i)\cdot\delta\vec{r}_i = 0$$

$$\sum_i \vec{F}_i\delta\vec{r}_i = \sum_k Q_k\delta q_k$$

$$\delta\vec{r}_i = \sum_{k=1}^{s}\frac{\partial\vec{r}_i}{\partial q_k}\delta q_k$$

$$\sum_k Q_k\delta q_k - \sum_i m_i\ddot{\vec{r}}_i\cdot\sum_k\frac{\partial\vec{r}_i}{\partial q_k}\delta q_k = \sum_k\left(Q_k - \sum_i m_i\ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_k}\right)\delta q_k = 0$$

δq_k 是独立的

$$\sum_i m_i\ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_k} = Q_k \quad (k=1,2,\cdots,s)$$

3. 应用前述关系式与对易关系，有：

$$Q_k = \sum_i m_i\ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_k} = \frac{d}{dt}\sum_i\left(m_i\dot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_k}\right) - \sum_i m_i\dot{\vec{r}}_i\cdot\frac{d}{dt}\left(\frac{\partial\vec{r}_i}{\partial q_k}\right)$$

$$\frac{\partial\vec{r}_i}{\partial q_k} = \frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_k} \qquad \frac{d}{dt}\left(\frac{\partial\vec{r}_i}{\partial q_k}\right) = \frac{\partial\dot{\vec{r}}_i}{\partial q_k}$$

$$= \frac{d}{dt}\sum_i\left(m_i\dot{\vec{r}}_i\cdot\frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_i}\right) - \sum_i m_i\dot{\vec{r}}_i\cdot\left(\frac{\partial\dot{\vec{r}}_i}{\partial q_k}\right)$$

$$\sum_{i=1}^{n} m_i\dot{\vec{r}}_i\cdot\frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_k} = \frac{\partial T}{\partial\dot{q}_k} \qquad \sum_{i=1}^{n} m_i\dot{\vec{r}}_i\cdot\left(\frac{\partial\dot{\vec{r}}_i}{\partial q_k}\right) = \frac{\partial T}{\partial q_k}$$

$$= \boxed{\frac{d}{dt}\frac{\partial T}{\partial\dot{q}} - \frac{\partial T}{\partial q_k} = Q_k, \quad k=1,2,\cdots,s.}$$

此即基本形式的拉格朗日方程，适用于理想，完整体系.